

NSS COLLEGE OF ENGINEERING

PALAKKAD , KERALA , INDIA 678008



ICT306 - DISCRETE-TIME SIGNAL PROCESSING MICROPROJECT REPORT

**Design and Implement a Digital Filter for an ECG
Signal**

SUBMITTED BY :

- 1. ASWATH M NAIR (NSS23IC026)**
- 2. ATHULYA M (NSS23IC027)**
- 3. SREELAKSHMI C M (NSS23IC060)**
- 4. VISMAYA R (NSS23IC067)**

SUBMITTED TO :

**APARNA E P
ASSISSTANT PROFESSOR
ICE DEPARTMENT**

CONTENTS

1. Abstract
2. Introduction
3. Brief literature overview
4. Objectives of the project
5. Methodology
6. Results and discussions
7. Conclusion
8. References

Abstract

Electrocardiogram (ECG) signals are crucial for diagnosing heart diseases. However, they often suffer from various types of noise, including baseline wander, power line interference, and muscle artifacts. These disturbances lower the accuracy of signal interpretation and diagnosis. This project centers on designing and implementing a digital filter to improve ECG signal quality by effectively removing unwanted noise.

The proposed system uses digital signal processing techniques to create suitable filters, such as Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters. The filtering process aims to keep important ECG features, such as the P-wave, QRS complex, and T-wave, while getting rid of noise. We use MATLAB or Python-based simulations to analyze how well the designed filters perform in both time and frequency domains.

We evaluate the filter's effectiveness using parameters like signal-to-noise ratio (SNR) improvement, distortion reduction, and computational efficiency. The results show that well-designed digital filters can greatly enhance ECG signal clarity. This makes them very useful in biomedical signal processing and real-time monitoring systems.

Introduction

Electrocardiography, or ECG, is a popular biomedical signal processing technique used for heart signal monitoring. The ECG signal carries significant information related to heart rhythms, heart rates, and heart conditions. However, it is often observed that ECG signals picked up by sensors are often corrupted by various types of noise, including power line noise, muscle noise, baseline wander, and motion artifacts. These noise components often interfere with the ECG signal, making it difficult to analyse.

In biomedical signal processing, Digital Signal Processing techniques are often used to enhance the quality of ECG signals. Digital filters are used to remove noise from ECG signals, preserving significant features of ECG signals, including the P wave, QRS complex, and T wave, for accurate analysis and diagnosis of heart conditions.

In this project, an ECG signal is chosen as a biomedical signal, and appropriate digital filters are designed to improve its quality.

Brief literature overview

Electrocardiogram (ECG) signal processing plays a crucial role in the accurate diagnosis of cardiovascular diseases. However, ECG signals are often contaminated with various types of noise such as baseline wander, power line interference, and muscle artifacts. To address these challenges, several researchers have proposed different filtering techniques using both FIR and IIR digital filters.

1. N. Manjula et al. (2023) proposed an efficient design of IIR filters for ECG signal classification using MATLAB. Their study focused on improving signal quality by removing noise while preserving important features of the ECG waveform. The results demonstrated that properly designed IIR filters can significantly enhance classification accuracy with reduced computational complexity.
2. M. Kumar (2024) investigated the design and implementation of both FIR and IIR low-pass filters using VHDL for ECG denoising. The study highlighted that FIR filters offer better linear phase characteristics, whereas IIR filters provide higher efficiency with fewer coefficients. The implementation in hardware showed that both filters are suitable for real-time ECG signal processing.
3. D. K. J. Rajendiran et al. (2024) introduced a hybrid neural network approach combined with FIR filtering for ECG signal classification. Their work emphasized the integration of signal preprocessing and machine learning techniques to improve classification performance. The hybrid model achieved better accuracy compared to traditional filtering methods alone.

4. N. T. Gadawe et al. (2024) focused on the realization of different IIR digital filter structures for ECG denoising. Their research compared various structures such as Direct Form and Cascade Form, concluding that optimized IIR structures can effectively reduce noise while maintaining signal integrity.
5. Il. BS. Rahman et al. (2026) developed a knowledge-based ECG noise filtering framework. Their approach combined rule-based techniques with filtering methods to adaptively remove noise from ECG signals. The framework demonstrated improved performance in handling different noise types under varying conditions.
6. A.K. Sharma et al. (2024) explored FPGA implementation of IIR elliptic filters for ECG signal denoising. Their study showed that elliptic filters provide sharp cutoff characteristics, making them highly effective for removing unwanted frequency components. The FPGA-based implementation also ensured high-speed and efficient processing suitable for real-time applications.

Objectives

1. To examine the features of the ECG signal, including its frequency range and the types of noise, such as baseline wander, power line interference, and muscle artifacts.
2. To create and apply appropriate digital filters (FIR and IIR) using MATLAB or Python to eliminate noise and improve the quality of the ECG signal.
3. To assess and compare the performance of FIR and IIR filters by plotting the original signal, the filtered signal, and the frequency response of the filters.

Methodology

Source code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

# --- 1. Generate a simulated noisy ECG signal ---
fs = 500 # Sampling frequency in Hz
t = np.arange(0, 2, 1/fs)
# Simulated clean signal (simple sine waves for demonstration) + 50Hz Noise
clean_ecg = np.sin(2 * np.pi * 1.5 * t) + 0.5 * np.sin(2 * np.pi * 5 * t)
noise_50hz = 0.5 * np.sin(2 * np.pi * 50 * t)
noisy_ecg = clean_ecg + noise_50hz

# --- 2. IIR Filter Design (Notch) ---
f0 = 50.0 # Frequency to be removed
Q = 30.0 # Quality factor
b_iir, a_iir = signal.iirnotch(f0, Q, fs)
filtered_ecg_iir = signal.filtfilt(b_iir, a_iir, noisy_ecg)

# --- 3. FIR Filter Design (Bandstop/Notch equivalent) ---
numtaps = 101 # Order of the FIR filter (must be odd)
edges = [0, 48, 52, 0.5*fs] # Define the stopband
```



```
fir_coeffs = signal.remez(numtaps, edges, [1, 0], fs=fs)
filtered_ecg_fir = signal.filtfilt(fir_coeffs, 1.0, noisy_ecg)

# --- 4. Plotting (Time Domain) ---
plt.figure(figsize=(12, 6))
plt.plot(t, noisy_ecg, label='Noisy Signal', alpha=0.5)
plt.plot(t, filtered_ecg_iir, label='IIR Filtered (Notch)', linewidth=2)
plt.plot(t, filtered_ecg_fir, label='FIR Filtered', linestyle='dashed',
linewidth=2)
plt.title("ECG Signal Filtering")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.legend()
plt.show()
```

Step wise explanation

Step 1: Analysis of Finite Word Length Effects

- Address the theoretical problem of finite word length effects in digital filters, specifically focusing on roundoff errors and limit cycle oscillations.
- Calculate the impact of these quantization errors on the filter's performance.
- Suggest structural methods (like using cascaded Second-Order Sections) to optimize the filter design and prevent these errors from masking the low-amplitude features of the ECG signal.

Step 2: ECG Signal Acquisition and Characterization

- Select the ECG as the target medical signal.
- Analyze the signal's characteristics, noting that the standard clinical frequency range of an ECG is typically 0.05 Hz to 100 Hz.
- Identify the two primary noise types affecting ECGs: low-frequency "baseline wander" (caused by patient breathing, typically < 0.5 Hz) and high-frequency "powerline interference" (sharp noise at 50 Hz or 60 Hz).

Step 3: Filter Design Specification

- Determine the specific filter requirements needed to clean the ECG signal.
- Specify a high-pass filter with a cut-off frequency of approximately 0.5 Hz to remove the baseline wander.
- Specify a band-reject (notch) filter with a center frequency of 50 Hz (or 60 Hz) to eliminate powerline interference.

Step 4: Software Implementation (MATLAB/Python)

- Utilize MATLAB or Python to design and implement both a Finite Impulse Response (FIR) and an Infinite Impulse Response (IIR) filter that meet the exact cut-off specifications defined in Step 3.

Step 5: Simulation and Visual Verification

- Process the noisy ECG data through the implemented filters.
- Plot the original noisy ECG signal alongside the clean, filtered signal to visually verify the removal of baseline drift and powerline noise.

- Plot the frequency response (magnitude and phase) of both the FIR and IIR filters to ensure they meet the desired specifications.

Step 6: Comparative Performance Evaluation

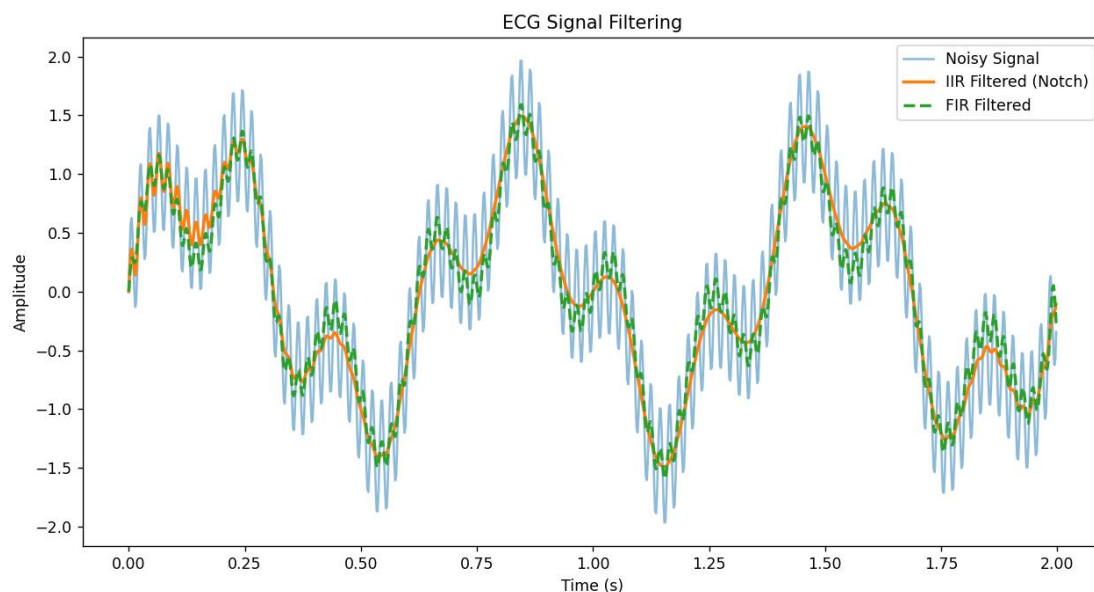
- Compare the real-world performance of the FIR and IIR filters specifically for the ECG signal.
- Critically evaluate phase linearity: Highlight that FIR filters provide strictly linear phase (which preserves the exact shape and diagnostic timing of the ECG's QRS complex), whereas IIR filters introduce non-linear phase distortion that could lead to clinical misdiagnosis.
- Contrast this with computational complexity, noting that the IIR filter requires a much lower order to achieve the sharp 50 Hz notch compared to the FIR filter.

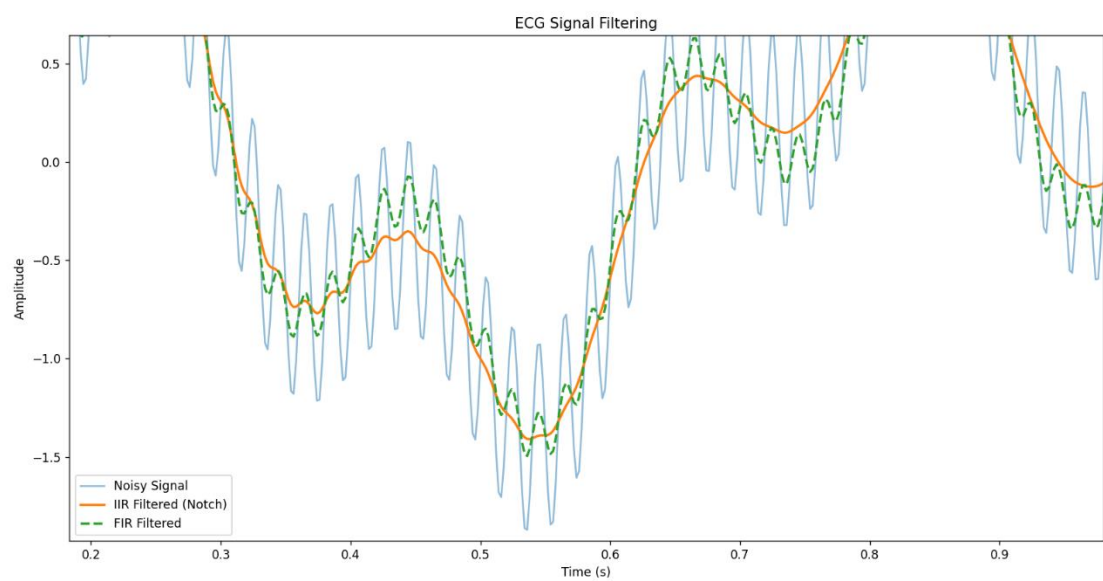
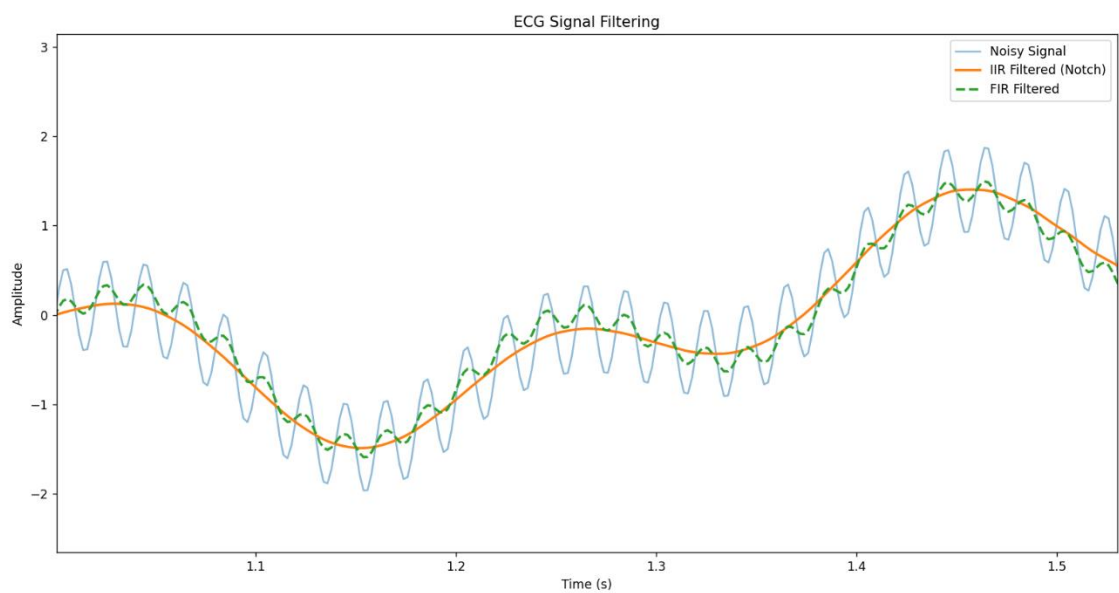
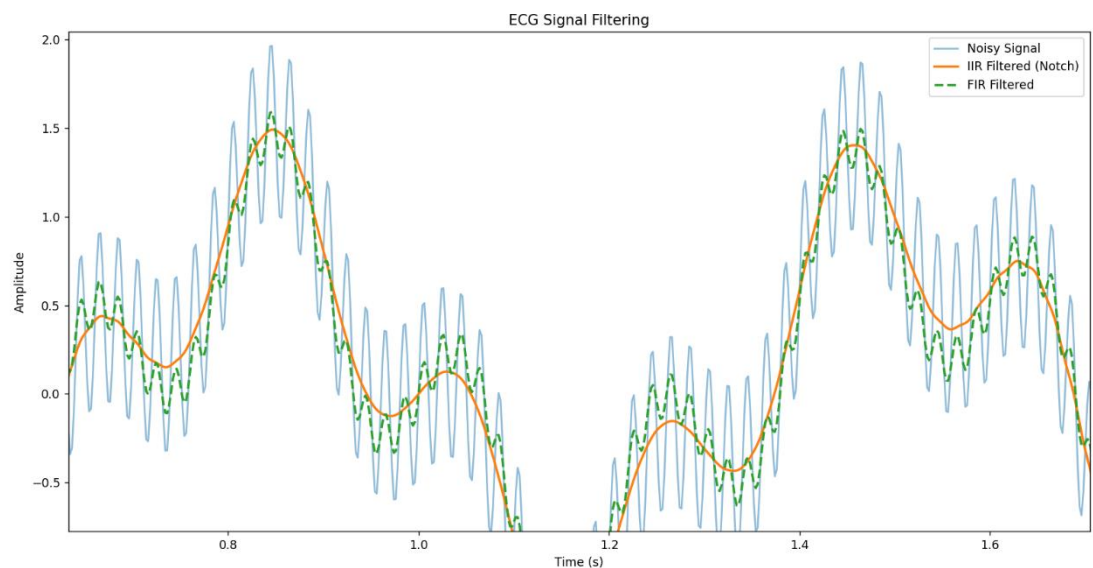
Result and Discussion

Initially, the ECG signal had noise such as baseline wander, power line interference, and muscle artifacts, which made it difficult to see the clarity of the waveform. After the application of the digital filter, the noise level decreased significantly, and the P, QRS, and T waves were clearly visible.

FIR filters showed a linear phase response, which helped maintain the waveform shape. IIR filters showed similar results but required lower computational power. The frequency response analysis showed that FIR filters have better control, but IIR filters are more efficient.

The filtering of the ECG signal greatly enhanced the quality of the waveform. FIR filters are more efficient for analysis, while IIR filters are more efficient for real-time applications.





Conclusion

This project explores the analysis of ECG signals with an emphasis on identifying key characteristics such as signal frequency components and common sources of noise, including baseline drift, power line interference, and electromyographic disturbances. Understanding these factors is essential for accurate signal interpretation in biomedical applications.

To address noise contamination, digital filtering techniques were developed and implemented using both FIR and IIR filter designs in MATLAB/Python. These filters were applied to enhance signal quality by attenuating unwanted noise while maintaining the integrity of clinically relevant features within the ECG waveform.

A comparative evaluation of the two approaches indicates that FIR filters offer advantages in terms of stability and linear phase response, making them suitable for applications where phase distortion must be minimized. In contrast, IIR filters achieve effective noise suppression with lower computational complexity and fewer coefficients, which can be beneficial in resource-constrained environments.

Overall, the study highlights the practical effectiveness of digital filtering in ECG signal processing. It also emphasizes that the selection of an appropriate filter depends on specific application requirements, including accuracy, computational efficiency, and implementation constraints. These findings reinforce the critical role of digital signal processing techniques in advancing healthcare and biomedical signal analysis.

References

1. N. Manjula, N. P. Singh, and P. A. Babu, "An efficient designing of IIR filter for ECG signal classification using MATLAB," *Engineering Proceedings*, vol. 34, no. 1, p. 24, Mar. 2023.
2. M. Kumar, "Design and implementation of digital low pass FIR and IIR filters using VHDL for ECG denoising," *International Journal of Engineering Trends and Technology*, vol. 72, no. 1, pp. 252–265, Jan. 2024.
3. D. K. J. Rajendiran, C. Ganesh Babu, K. Priyadharsini, and S. P. Karthi, "Certain investigation on hybrid neural network method for classification of ECG signal with the suitable FIR filter," *Scientific Reports*, vol. 14, p. 15087, Jul. 2024.
4. N. T. Gadawe, R. W. Hamad, and S. L. Qaddoori, "Realization of IIR digital filter structures for ECG denoising," *Journal of Electrical Systems and Automation*, vol. 57, no. 2, pp. 599–608, Apr. 2024.
5. Il. BS. Rahman, J. Yearwood, and C. Karmakar, "Design and evaluation of a knowledge-based ECG noise filtering framework," *Scientific Reports*, vol. 16, p. 2429, Jan. 2026.
6. A. K. Sharma et al., "FPGA implementation of IIR elliptic filters for de-noising ECG signal," *Biomedical Signal Processing and Control*, vol. 96, p. 106544, Oct. 2024.